BYoung Physical Therapy

Return to Sport Assessment

Patient: Alex Walker **Surgical Side:** R **Date:** April 02, 2026

88 RTS Score
Ready

Return to Sport Profile

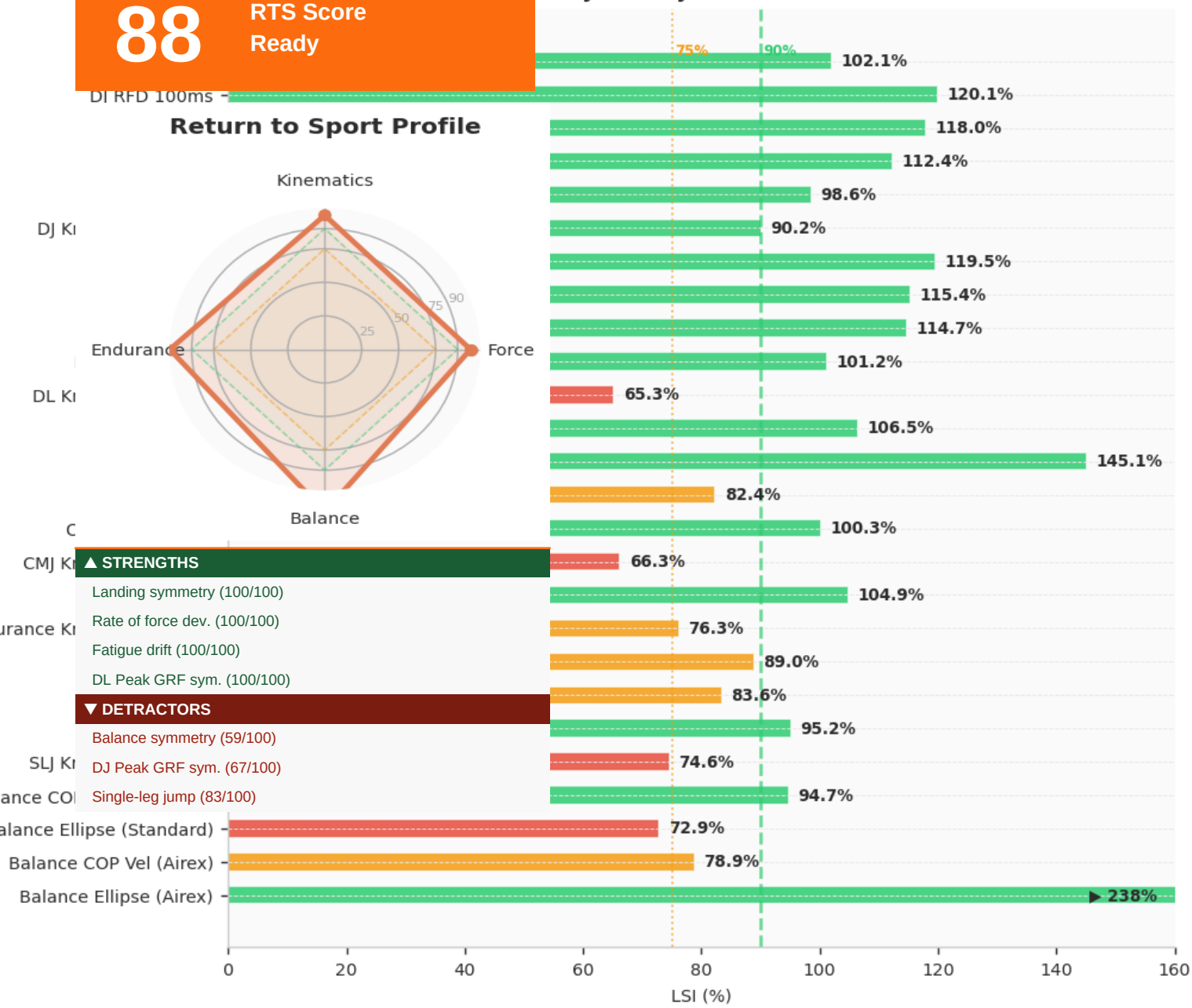
▲ STRENGTHS

- Landing symmetry (100/100)
- Rate of force dev. (100/100)
- Fatigue drift (100/100)
- DL Peak GRF sym. (100/100)

▼ DETRACTORS

- Balance symmetry (59/100)
- DJ Peak GRF sym. (67/100)
- Single-leg jump (83/100)

nb Symmetry Index — All Tests



CLINICAL FLAGS

- DJ Ankle Dorsiflexion: Δ -8.2° — exceeds 6° threshold
- DL Knee Ext Moment: LSI 65.3% — significant asymmetry
- CMJ Knee Ext Moment: LSI 66.3% — significant asymmetry
- SLJ Knee Ext Moment: LSI 74.6% — significant asymmetry
- Balance Ellipse (Standard): LSI 72.9% — significant asymmetry

● ≥90% Symmetric ● ≥75% Progressing ● <75% Deficit LSI = Surgical ÷ Non-Surgical × 100 | Δ = Surgical – Non-Surgical (absolute difference; used for small-angle kinematics)

DROP JUMP

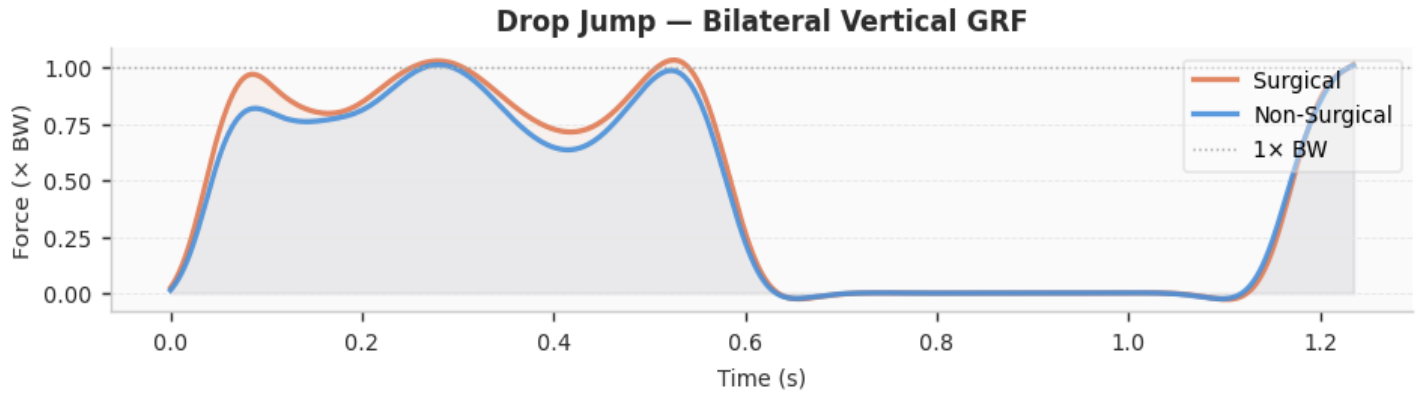
Metric	Surgical	Non-Surg	LSI / Δ
Peak GRF (N)	874.8	856.9	102.1%
Impulse (N·s)	468.3	438.6	106.8%
RFD 100ms (N/s)	9317.4	7759.8	120.1%
Loading Rate 50ms (N/s)	11294.7	9568.7	118.0%
Landing Sym 200ms (%) *(bilateral)*	—	—	112.4%
Impact Transient (N)	584.7	489.0	119.6%
Time-to-Peak (ms)	526.0	279.0	—
RSI	1.540	—	—
Contact Time (s)	0.623	—	—
Peak Knee Flexion (°)	95.7	97.1	98.6%
Peak Knee Valgus (°)	4.2	1.7	Δ +2.5°
Peak Tibial IR (°)	0.5	1.7	Δ -1.2°
Tibial Rot Excursion (°)	5.2	7.1	Δ -1.9°
Peak Hip Flexion (°)	—	—	—
Peak Hip Adduction (°)	-1.7	-3.3	Δ +1.6°
Peak Hip IR (°)	—	—	—
Peak Ankle Dorsiflexion (°)	85.7	93.9	Δ -8.2°
Peak Knee Ext Moment (Nm)	116.1	128.8	90.2%
AP COP Excursion (mm)	351.2	457.8	—

CLINICAL INTERPRETATION

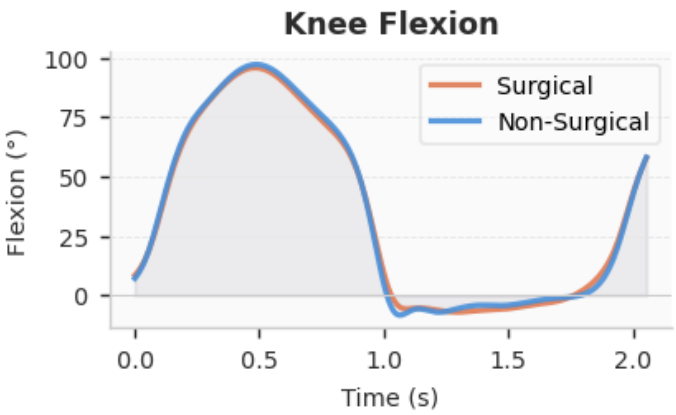
RSI 1.54 (✓ meets RTS target ≥ 1.30). Reactive capacity is adequate.

Landing symmetry 112% LSI: Surgical limb loading 12% more than non-surgical — protective offloading of NS side detected.

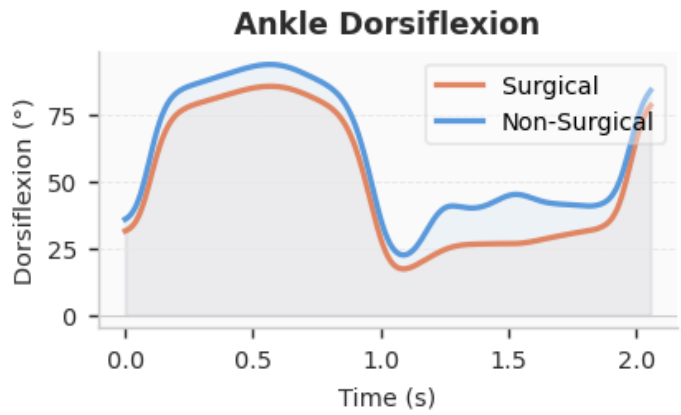
Targets: RSI ≥ 1.30 · Landing LSI 90–110% · Load rate LSI ≥ 90% · Knee flexion ≥ 60° · Valgus < 5° · Tibial IR < 10°.



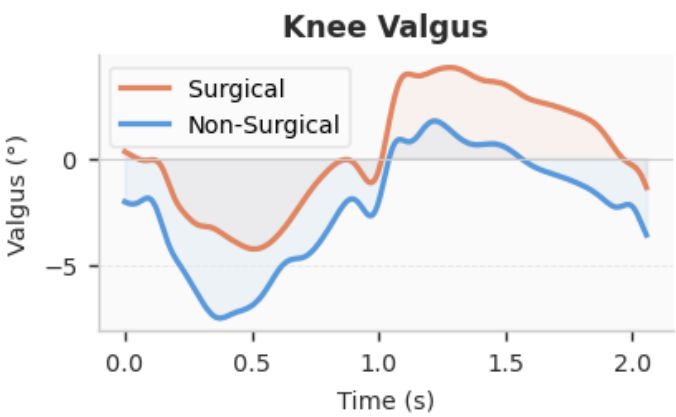
Bilateral GRF (normalised to BW). Orange = surgical, Blue = non-surgical.



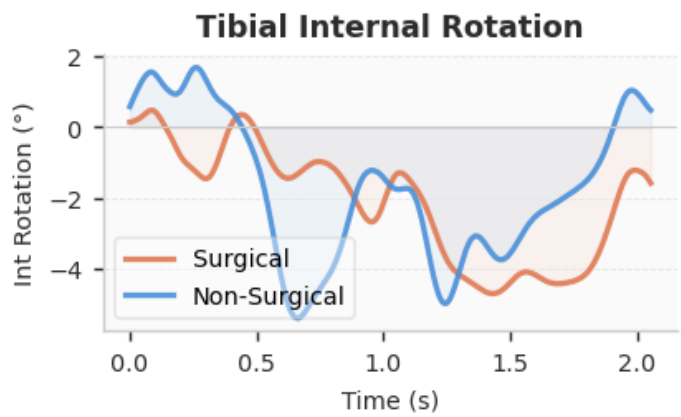
Knee flexion. Target ≥ 60° at contact. Reduced depth = stiff landing strategy.



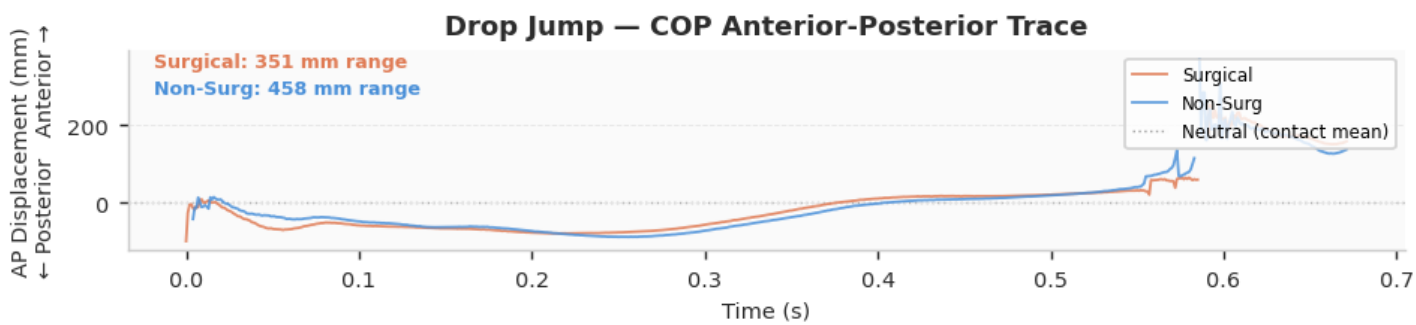
Ankle dorsiflexion. Restricted dorsiflexion → shifts load to quadriceps / increases ACL stress.



Knee valgus. Target < 5°. Elevated valgus = medial collapse / ACL re-injury risk.



Tibial internal rotation. IR > 10° at landing = increased ACL rotational load.



AP COP displacement (zero-centred). Smaller excursion on surgical limb = restricted fore-aft weight transfer / guarded landing strategy.

ABOUT THIS TEST

The Drop Jump is a reactive bilateral landing task performed from a standard box height. The athlete drops off the box, absorbs the impact with both limbs simultaneously, then immediately jumps for maximum height. It tests the athlete's ability to rapidly store and release elastic energy (reactive strength) and to absorb high-impact loads symmetrically. Key concerns after ACL reconstruction include stiff-knee landing (reduced flexion, elevated impact transient), medial knee collapse (valgus), and side-to-side force asymmetries that indicate ongoing limb offloading. The Reactive Strength Index (RSI = jump height ÷ contact time) summarises the explosive reactive capacity of the neuromuscular system.

DROP LANDING

Metric	Surgical	Non-Surg	LSI / Δ
Peak GRF (N)	789.3	660.4	119.5%
Impulse (N·s)	484.0	447.5	108.1%
Loading Rate 50ms (N/s)	11640.7	10087.3	115.4%
Impact Transient (N)	603.5	526.3	114.7%
Peak Knee Flexion (°)	103.8	102.5	101.2%
Peak Knee Valgus (°)	1.2	-0.5	Δ +1.7°
Peak Tibial IR (°)	0.7	2.2	Δ -1.5°
Tibial Rot Excursion (°)	3.8	3.0	Δ +0.9°
Peak Hip Flexion (°)	—	—	—
Peak Hip Adduction (°)	-3.6	-8.1	Δ +4.5°
Peak Hip IR (°)	—	—	—
Peak Ankle Dorsiflexion (°)	82.2	88.2	Δ -6.0°
Peak Knee Ext Moment (Nm)	62.7	96.0	65.3%
AP COP Excursion (mm)	147.7	167.1	—

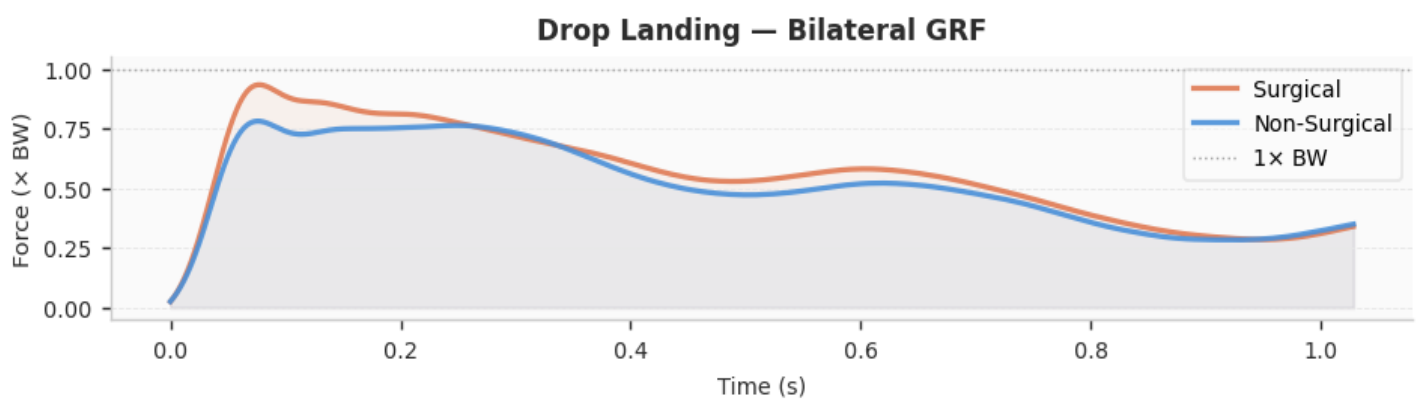
CLINICAL INTERPRETATION

Absorptive landing — assesses shock attenuation without a re-jump. Lower, more gradual force = better control.

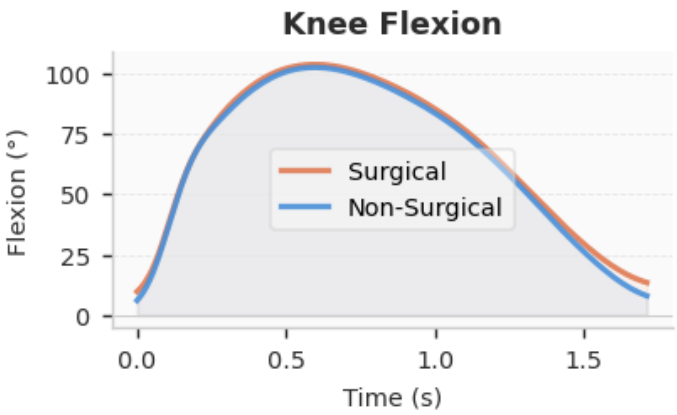
Load rate LSI 115%: Surgical limb loading rate 15% higher — NS side protective offloading.

Impact transient LSI 115%. Target: < 2× BW peak, LSI 90–110%, no sharp spike.

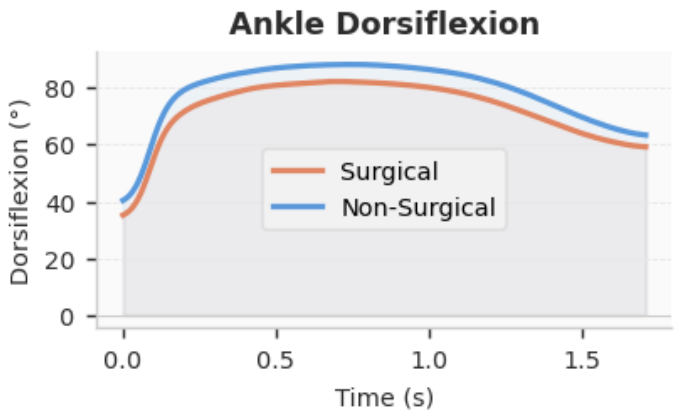
Targets: Peak GRF < 3× BW · Load rate LSI 90–110% · Knee flexion ≥ 60° · Valgus < 5° · Tibial IR < 10°.



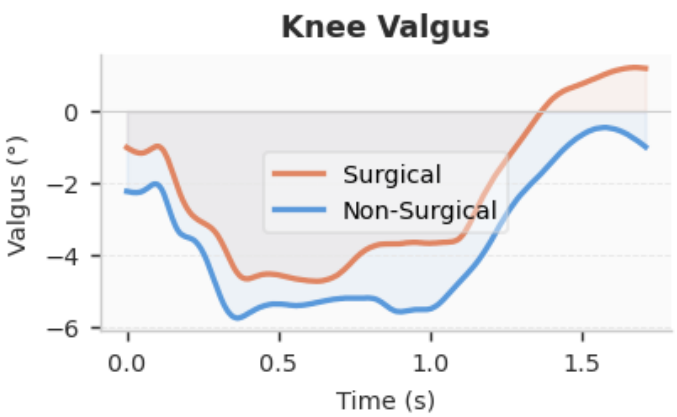
Bilateral GRF. Smooth progressive rise = good shock absorption. Sharp impact spike = stiff landing.



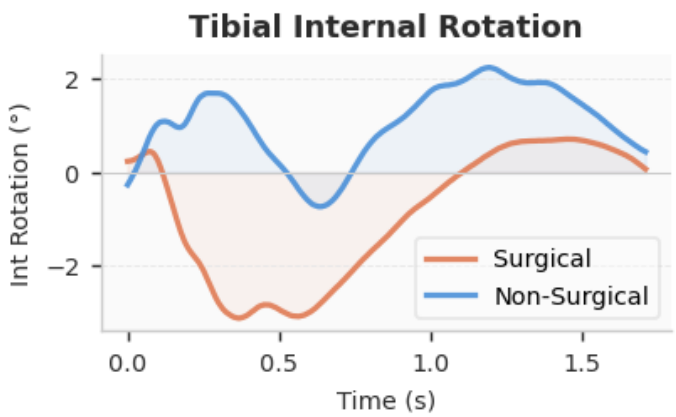
Knee flexion at landing. Reduced depth = stiff strategy / quad inhibition.



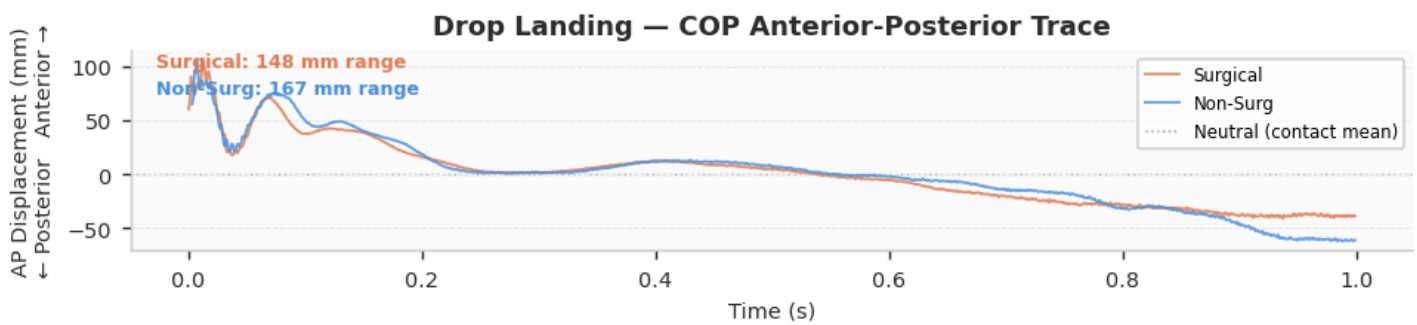
Ankle dorsiflexion. Restricted ROM limits knee flexion depth and increases ACL load.



Knee valgus at landing. > 8° surgical = elevated ACL stress.



Tibial IR at landing. Combined with valgus = high ACL injury-risk pattern.



AP COP (zero-centred). Posterior bias on surgical side = heel-dominant / reduced forefoot loading.

ABOUT THIS TEST

The Drop Landing isolates the deceleration phase of a jump: the athlete drops from a box and sticks the landing bilaterally without a subsequent jump. This removes the propulsive demand and focuses entirely on the athlete's ability to safely absorb and dissipate ground reaction forces. After ACL reconstruction, common compensations include a stiff-knee strategy (reduced flexion, sharp impact spike), excessive valgus collapse, and asymmetric weight acceptance—all of which increase mechanical stress at the reconstructed ligament. A smooth, progressive force curve with symmetric knee flexion ≥ 60° and valgus < 8° is the clinical target. Loading rate and impact transient LSI reflect side-to-side shock absorption capacity.

MAX VERTICAL JUMP (CMJ)

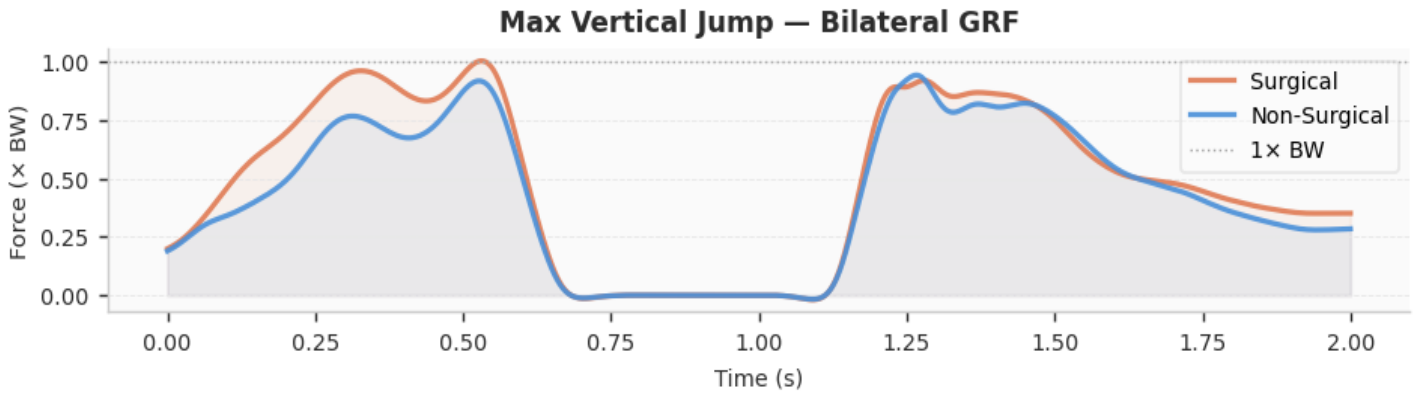
Metric	Surgical	Non-Surg	LSI / Δ
Peak Concentric GRF (N)	847.6	796.1	106.5%
Propulsion Impulse (N·s)	500.8	345.2	145.1%
Peak Force Surg / NS (N)	518.6	629.4	82.4%
Propulsion LSI (%) *(bilateral)*	—	—	145.1%
Jump Height	18.4	—	—
Flight Time (s)	0.442	—	—
Unweighting Impulse Surg/NS	262.5	290.6	—
Peak Knee Flexion (°)	104.0	103.7	100.3%
Peak Knee Valgus (°)	4.4	1.3	Δ +3.0°
Peak Tibial IR (°)	2.9	6.2	Δ -3.3°
Tibial Rot Excursion (°)	7.4	10.6	Δ -3.2°
Peak Hip Flexion (°)	—	—	—
Peak Hip Adduction (°)	-0.6	-2.7	Δ +2.1°
Peak Hip IR (°)	—	—	—
Peak Ankle Dorsiflexion (°)	86.5	92.8	Δ -6.3°
Peak Knee Ext Moment (Nm)	73.7	111.1	66.3%
AP COP Excursion (mm)	367.6	415.1	—

CLINICAL INTERPRETATION

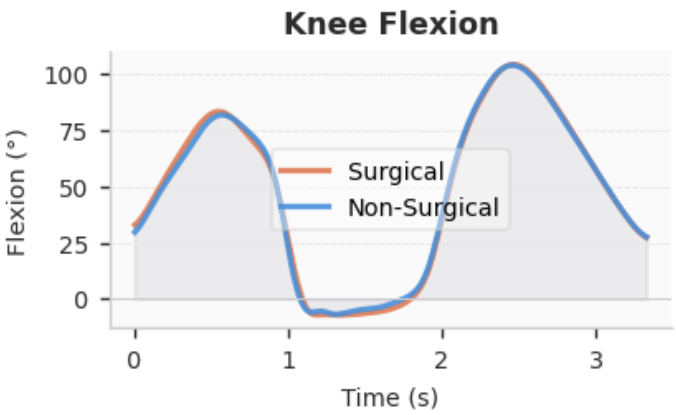
Bilateral CMJ — sensitive to subtle propulsion deficits masked at lower intensities.

Peak force LSI 82%: Surgical limb generating 18% less force — quad/hip extensor deficit; strength work required.

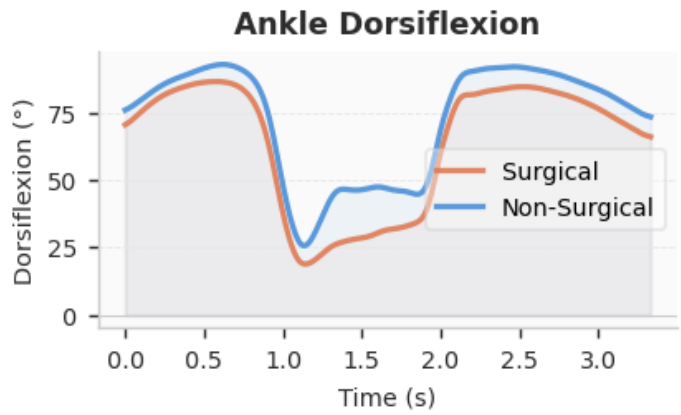
Propulsion LSI 145%. Targets: ≥ 90% propulsion, ≥ 90% peak force, matched unweighting impulse.



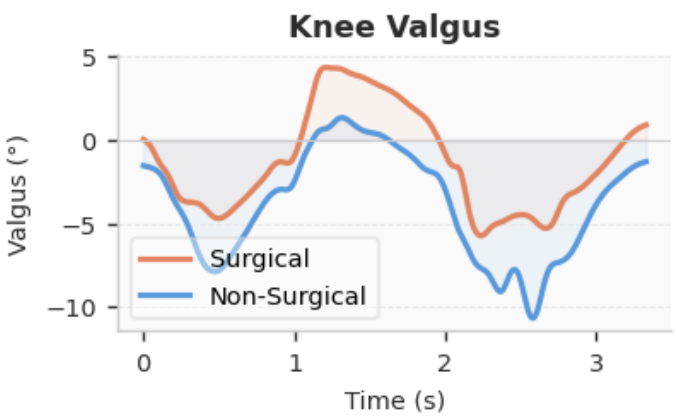
Full CMJ: unweighting → eccentric load → propulsion → flight. Propulsive peaks should be symmetric.



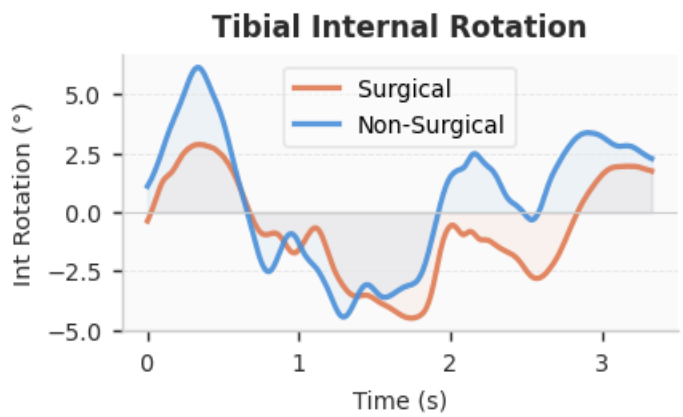
Knee flexion depth. Asymmetric countermovement depth = compensatory loading strategy.



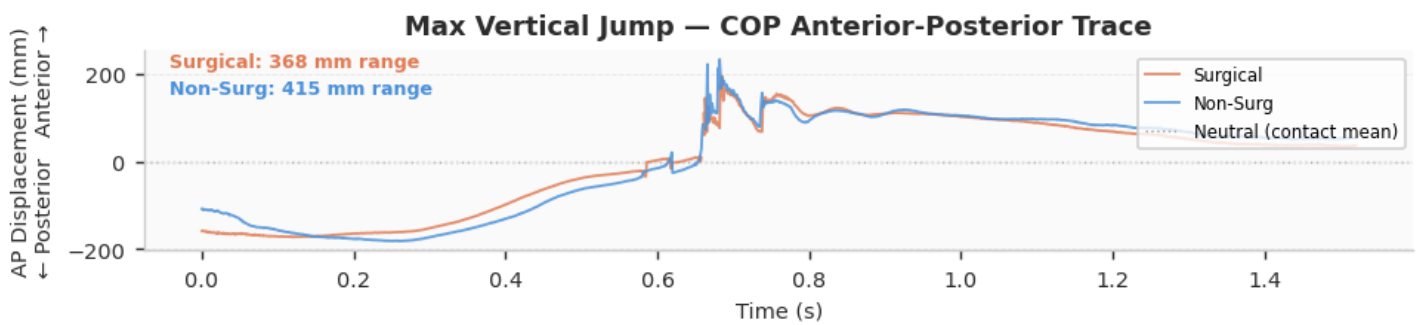
Ankle dorsiflexion. Restricted ROM reduces jump height and shifts load away from the knee.



Knee valgus during propulsion. Dynamic valgus under load = ACL stress.



Tibial IR during propulsion. Coupled with valgus = high re-injury risk pattern.



AP COP displacement during CMJ (zero-centred). Anterior shift during propulsion drives jump height — asymmetric or reduced excursion on surgical side = limb offloading.

ABOUT THIS TEST

The Countermovement Jump (CMJ) is the gold-standard bilateral power test. The athlete begins in quiet stance, rapidly descends into a squat (countermovement), then drives upward for maximum jump height. The full GRF waveform reveals each phase: unweighting, eccentric loading, propulsion, and flight. After ACL reconstruction, typical deficits include an asymmetric countermovement depth (surgical limb offloading during the eccentric phase), reduced propulsive impulse, and side-to-side jump height asymmetry. Propulsive LSI < 90% is a common criterion for RTS decisions. Kinematic analysis identifies compensatory strategies—valgus, reduced knee flexion, and tibial rotation—that indicate the surgical limb is not being loaded with full confidence or mechanical efficiency.

ENDURANCE SQUAT (30s)

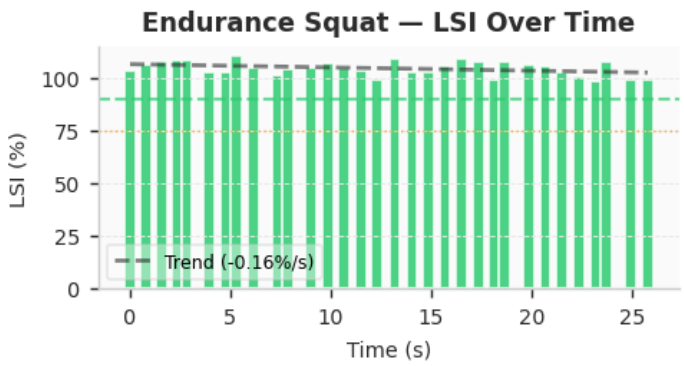
Metric	Surgical	Non-Surg	LSI / Δ
Mean Peak Force LSI (%) *(bilateral)*	—	—	104.9%
LSI First 10s (%) *(bilateral)*	—	—	106.1%
LSI Last 10s (%) *(bilateral)*	—	—	103.1%
Fatigue Drift (Δ%)	-2.9	—	—
Total Reps Detected	32.0	—	—
Mean Peak Force Surg (N)	872.4	—	—
Mean Peak Force NS (N)	831.8	—	—
AP COP Excursion — Surg / NS (mm)	196.4	218.3	—
Peak Knee Ext Moment (Nm)	93.0	121.9	76.3%

CLINICAL INTERPRETATION

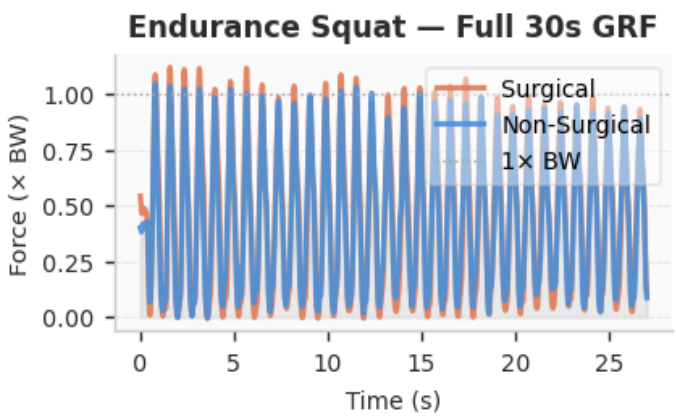
30s bilateral squat — reveals neuromuscular fatigue-induced asymmetry missed by single-rep testing.

Fatigue drift -2.9%: ✓ Symmetry maintained under fatigue.

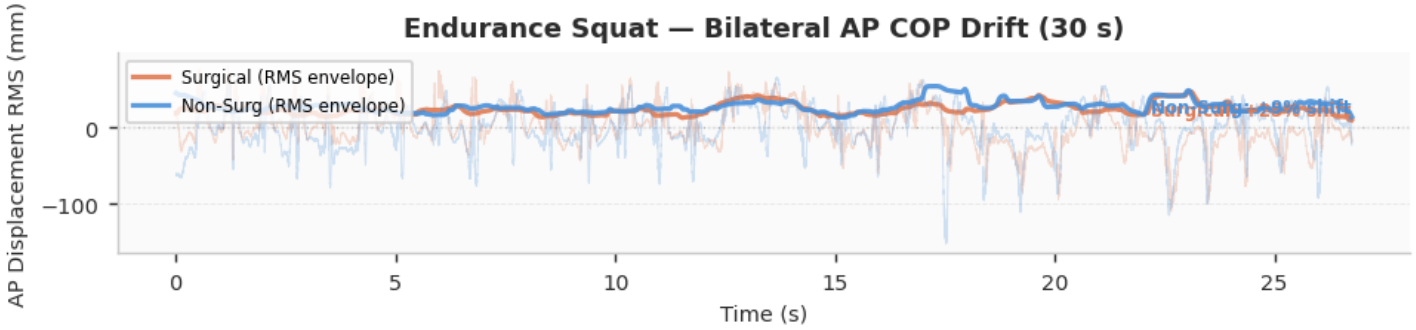
Targets: Mean LSI ≥ 90% · Fatigue drift < 5% · No progressive offloading across 30s.



Per-rep LSI across 30s. Declining trend = neuromuscular fatigue breakdown.



Full 30s GRF trace. Force magnitude and rhythm across the set.



Bilateral AP COP drift across 30s (zero-centred, 1s RMS envelope). Surgical-side amplitude shrinking or trending posterior = offloading under fatigue. Diverging traces = progressive asymmetry with sustained effort.

ABOUT THIS TEST

The 30-Second Endurance Squat assesses neuromuscular fatigue and limb symmetry under sustained bilateral loading. The athlete performs continuous bilateral squats to a standardised depth for a full 30 seconds. Unlike single-repetition tests, this task reveals how well the neuromuscular system maintains symmetry as it fatigues. Athletes post-ACL reconstruction frequently show an initial LSI within acceptable range that deteriorates progressively across the set as the nervous system disinhibits protective guarding. The COP drift trace exposes this compensation: surgical-limb weight transfer decreases, traces diverge, and loading shifts toward the uninvolved side. This test is particularly sensitive to residual quadriceps inhibition that is not apparent during brief, high-energy tasks.

SINGLE-LEG VERTICAL JUMP

Metric	Surgical	Non-Surg	LSI / Δ
Jump Height	9.9	11.2	89.0%
Peak GRF (N)	1284.1	1536.7	83.6%
Propulsion Impulse (N·s)	695.6	730.5	95.2%
RFD 100ms (N/s)	1505.4	—	—
Flight Time (s)	0.2	0.3	—
Contact Time (s)	0.8	0.8	—
Peak Knee Flexion (°)	50.6	59.2	—
Peak Knee Valgus (°)	2.2	-0.7	Δ +2.9°
Peak Tibial IR (°)	1.6	3.0	Δ -1.4°
Peak Ankle Dorsiflexion (°)	85.7	95.3	Δ -9.6°
Peak Hip Adduction (°)	10.5	25.3	Δ -14.7°
Peak Knee Ext Moment (Nm)	98.2	131.6	74.6%
AP COP Excursion (mm)	287.5	238.5	—

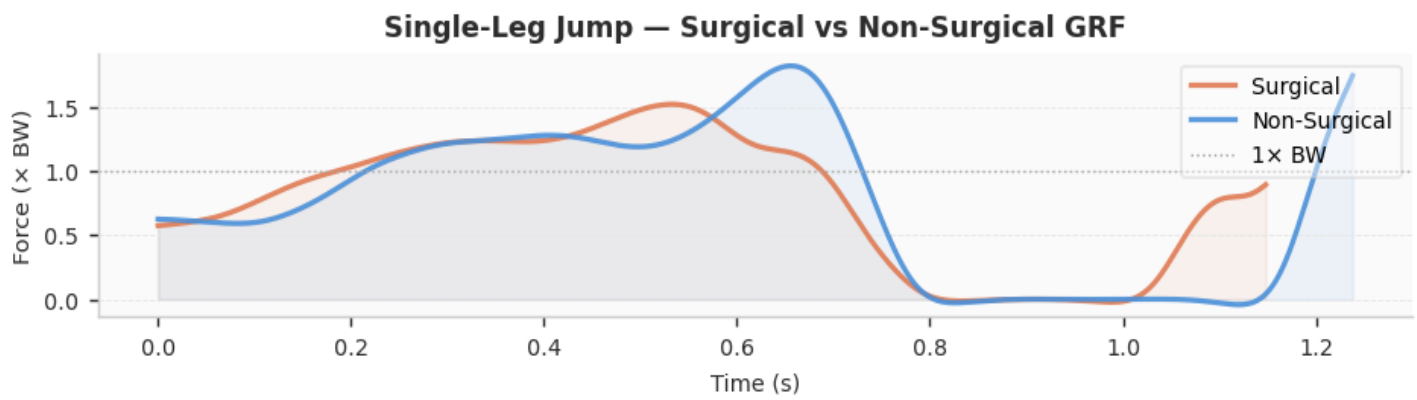
CLINICAL INTERPRETATION

Highest-demand test — isolates each limb, removing bilateral compensation. Strongest re-injury risk predictor.

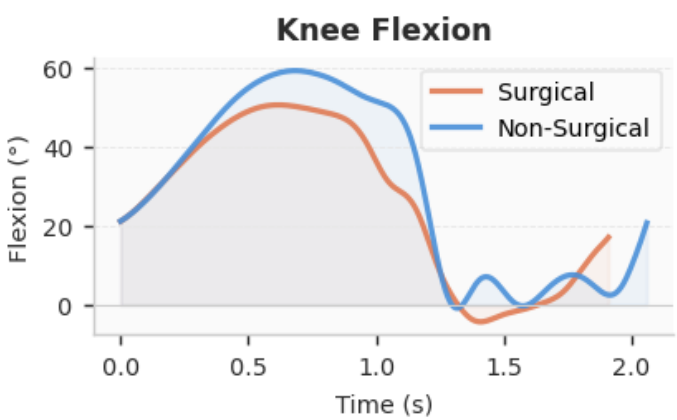
Jump height LSI 89%: ■ 11% below non-surgical. Continued plyometric loading required.

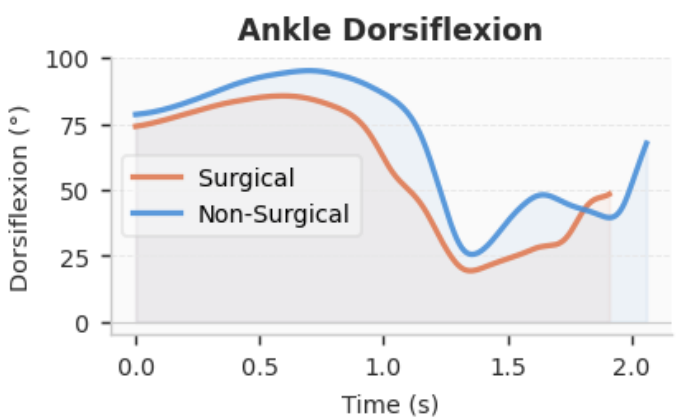
Peak force LSI 84%: Force deficit on surgical side — prioritise quad/hip extensor strengthening.

Targets: Jump height LSI ≥ 90% · Peak GRF LSI ≥ 90% · Valgus < 5° · Tibial IR < 10° on surgical limb.



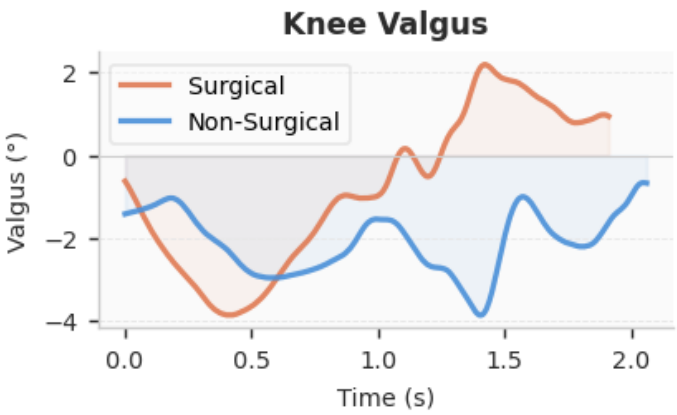
Per-limb GRF (separate trials). Propulsive peak height and shape should match.

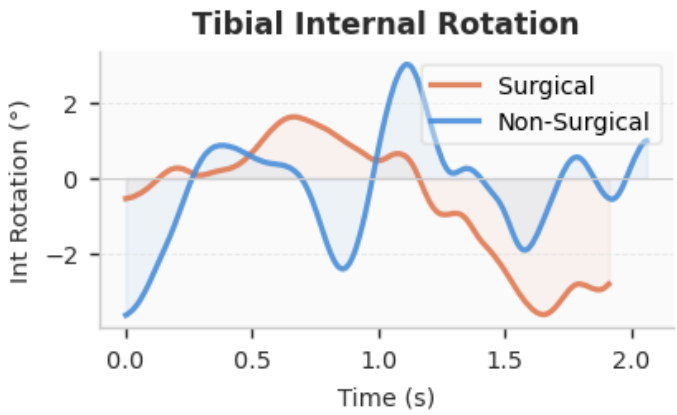




Knee flexion per limb. Reduced surgical depth = power deficit or guarded strategy.

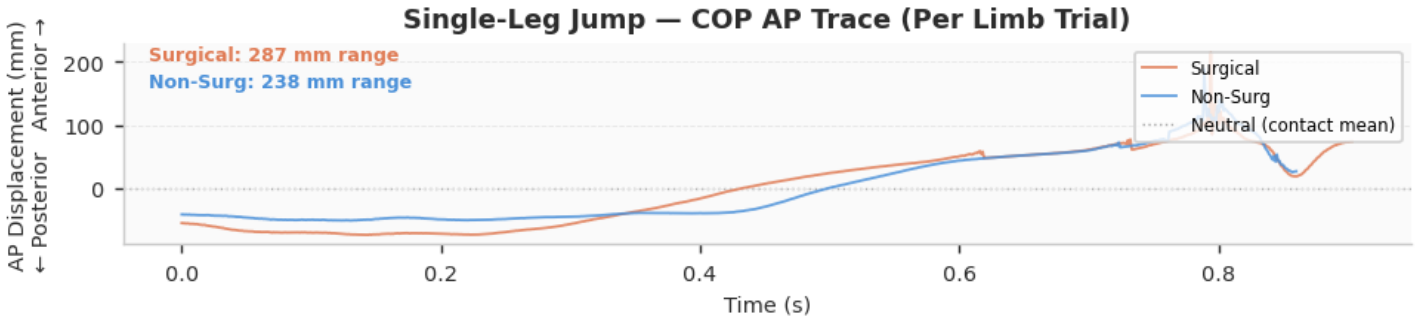
Ankle dorsiflexion per limb. Restricted ROM = reduced propulsive range / increased ACL load.





Knee valgus per limb. Elevated surgical valgus under single-leg load = re-injury risk.

Tibial IR per limb. Surgical-side IR > 10° = high rotational ACL stress pattern.



AP COP displacement (zero-centred, per limb trial). Reduced anterior excursion on surgical side = guarded takeoff / limited push-off.

ABOUT THIS TEST

The Single-Leg Vertical Jump isolates each limb independently, removing the ability of the uninvolved leg to compensate. The athlete performs a maximal countermovement jump landing on the same leg. Jump height LSI (surgical vs. non-surgical) is among the most sensitive power-based RTS criteria—a threshold of ≥ 90% is widely cited in ACL literature. Single-leg testing also reveals subtle compensations masked in bilateral tasks: reduced propulsive depth, hip drop (Trendelenburg pattern), and increased tibial rotation are common markers of residual neuromuscular asymmetry. Pelvis drop depth quantifies hip abductor control on the stance limb, which is an independent predictor of re-injury risk following return to sport.

BALANCE — STANDARD (EYES CLOSED, FIRM SURFACE)

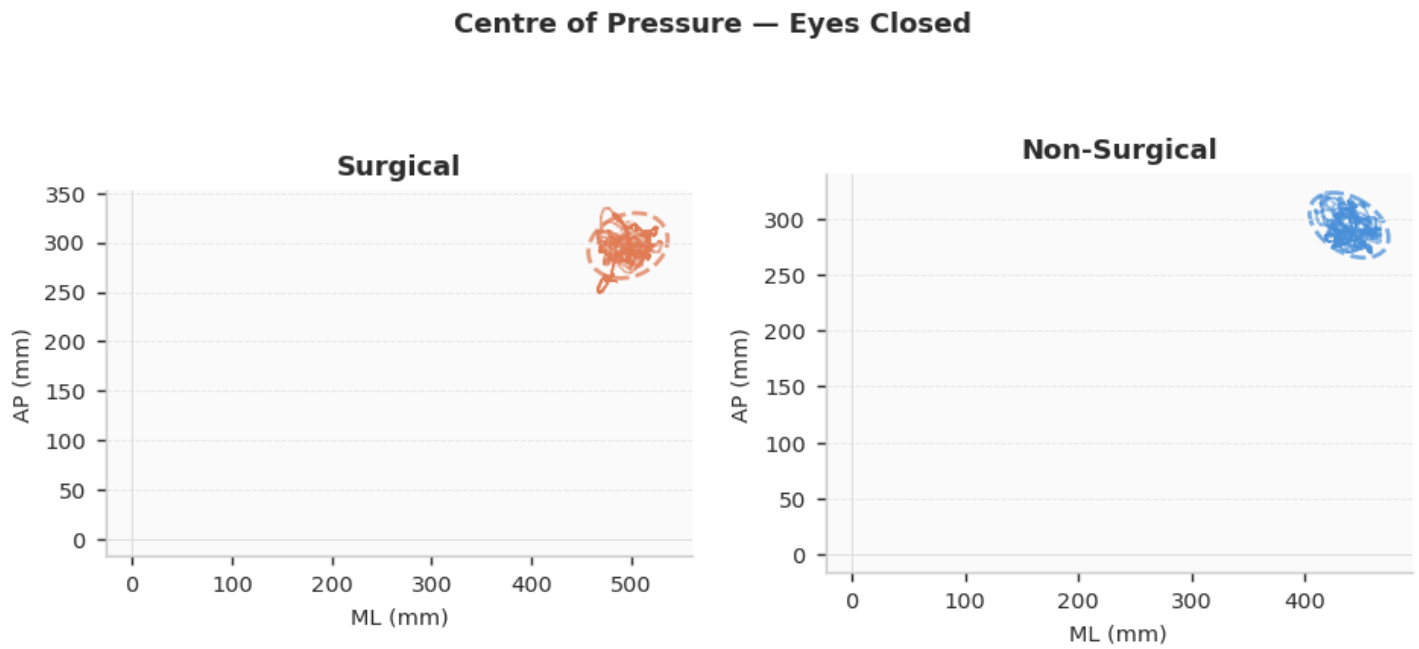
Metric	Surgical	Non-Surg	LSI / Δ
COP Mean Velocity (mm/s)	454.5	430.6	94.7%
COP Ellipse Area (mm²)	4036.6	2943.3	72.9%
Total Excursion (mm)	6362.9	6027.5	—
AP Range (mm)	85.9	54.3	—
ML Range (mm)	65.4	58.3	—
RMS Displacement (mm)	21.1	18.6	—

CLINICAL INTERPRETATION

Eyes-closed single-leg stance on firm surface. COP velocity is the most sensitive metric — higher speed = more corrective activity = reduced proprioception. Target: < 12 mm/s.

Surgical 455 mm/s vs Non-Surgical 431 mm/s — Surgical limb less stable. Proprioceptive retraining indicated.

COP Velocity LSI 95% (NS÷Surg×100 — lower = better). Normative target: < 12 mm/s. LSI target ≥ 90%.



COP trajectory — eyes closed. Tighter ellipse = better joint position sense. 95% confidence ellipse shown.

ABOUT THIS TEST

The Balance / Proprioception test assesses joint position sense and neuromuscular control under sensory challenge. The athlete stands on a single leg with eyes closed, removing visual compensation and isolating the mechanoreceptor feedback system of the ankle and knee. Center of Pressure (COP) velocity is the most sensitive metric: a higher velocity indicates greater corrective activity, meaning the neuromuscular system is working harder to maintain stability. After ACL reconstruction, deficits in mechanoreceptor density within the graft and capsular tissue result in elevated COP velocity and a larger sway ellipse on the surgical limb. The Airex pad condition adds an unstable surface to further stress the proprioceptive system, revealing deficits that may be masked on firm ground. Symmetry between limbs (LSI ≥ 85–90%) is the primary RTS criterion for this test.

BALANCE — AIREX PAD (EYES CLOSED, UNSTABLE SURFACE)

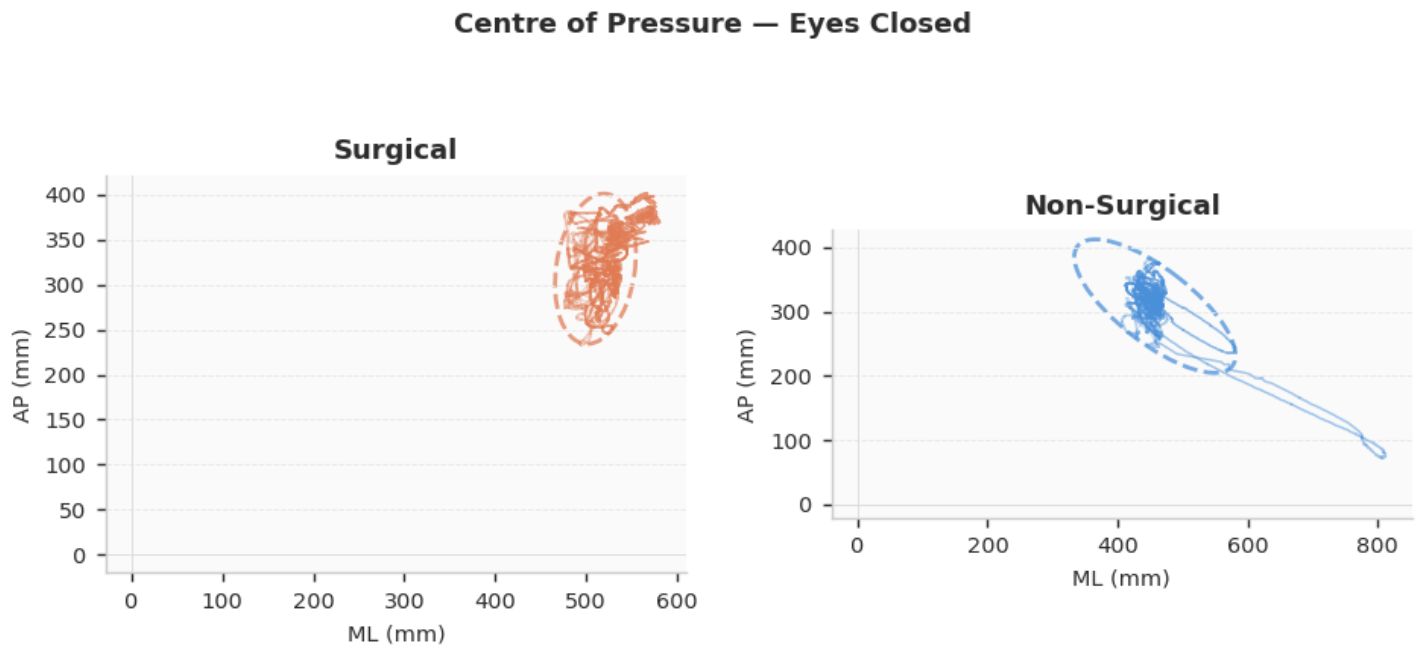
Metric	Surgical	Non-Surg	LSI / Δ
COP Mean Velocity (mm/s)	726.9	573.7	78.9%
COP Ellipse Area (mm²)	11439.1	27205.6	237.8%
Total Excursion (mm)	10163.4	8031.6	—
AP Range (mm)	168.7	311.1	—
ML Range (mm)	105.6	406.1	—
RMS Displacement (mm)	38.7	66.1	—

CLINICAL INTERPRETATION

Eyes-closed single-leg stance on Airex foam pad. Higher COP values are expected — the unstable surface challenges proprioceptive and neuromuscular systems. Symmetry between limbs (LSI ≥ 85%) is the key target.

Surgical 727 mm/s vs Non-Surgical 574 mm/s — Surgical limb less stable. Proprioceptive retraining indicated.

COP Velocity LSI 79% (NS÷Surg×100 — lower = better). Normative target: < 50 mm/s. LSI target ≥ 90%.



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